Responsibility-Based Socially Compatible Driving Behavior Modeling Verified by Hierarchical Multi-Agent Inverse Reinforcement Learning

Tingjun Li, Nan Xu, Shuo Feng, *Member, IEEE*, Hassan Askari, Bruno Henrique Groenner Barbosa, and Konghui Guo

***Abstract*— Autonomous vehicles (AVs) offer a promising glimpse into a future where transportation is smarter, safer, and more streamlined. Nevertheless, as AVs continue to interact with conventional vehicles (CVs), the potential for increased complexities and challenges cannot be overlooked, such as the frozen robot problem. This study proposes a regret-based model for motion planning responsibilities, encompassing self-respect and courtesy for conflicting personal interests. By incorporating these reciprocal responsibilities, socially compatible driving behaviors are promoted, and uncertainties in behavior are also reduced. A Self-Respect-Courtesy (SR-C) plane is further introduced, illustrating the interaction intensity and tendency. To navigate the trade-offs of responsibilities in varying situations, the concept of environmental niche is provided. Niches help to characterize the outcomes of specific actions with the resulting conditions to fulfill responsibilities. Finally, a hierarchical multi-agent inverse reinforcement learning algorithm is designed to calibrate the proposed model with NGSIM highway lane-changing cases. We found that the proposed model can significantly improve the calibration results and reduce the predictions error of mandatory lane changes by up to 20%. Moreover, the cross-entropy error also significantly decreases in a stable stage, indicating that responsible actions can safely reduce the behavior uncertainties of interactions. Our research revealed that drivers prioritize courtesy responsibility in discretionary lane changes with more consistency, whereas their self-respect preferences are stronger but show more variability in mandatory lane changes. These findings provide valuable insights into the underlying mechanism of interactions.**

***Index Terms*— Driving behavior modeling, responsibility, ethical motion planning, social compatibility, multi-agent inverse reinforcement learning, game theory.**

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Tingjun Li, Nan Xu, and Konghui Guo are with the National Key Laboratory of Automotive Chassis Integration and Bionics, Jilin University, Changchun, Jilin 130025, China (e-mail: litj21@mails.jlu.edu.cn; xunan@jlu.edu.cn; guokh@jlu.edu.cn).

Shuo Feng is with the Department of Automation, Beijing National Research Center for Information Science and Technology, Tsinghua University, Beijing 100084, China (e-mail: fshuo@tsinghua.edu.cn).

Hassan Askari is with the Department of Engineering, Brock University, St. Catharines, ON L2S 3A1, Canada (e-mail: haskari@brocku.ca).

Bruno Henrique Groenner Barbosa is with the Department of Automatics, Federal University of Lavras, Lavras, Minas Gerais 37200-000, Brazil (e-mail: brunohb@ufla.br).

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I. Introduction

UNDERSTANDING the social interactions among human drivers can help autonomous vehicles (AVs) act more responsibly.

Despite technological advancements, AVs still face challenges in achieving responsibility-based social compatibility, necessitating cautious driving practices for safe operation. In addition, the AVs may decelerate or take extreme evasive action if other road users behave with too much uncertainty, known as the "freezing robot problem" [1]. Unfortunately, the overly cautious behavior of AVs not only often leads to traffic congestion, particularly at intersections but also exposes them to risk from aggressive conventional vehicles (CVs). A study examining California traffic accidents found that rear-end collisions were significantly more prevalent in accidents involving AVs, accounting for 69% in 2022 [2]. An assumption for this difference is that drivers of the CVs are not accustomed to the driving style of AVs in the convoy [3]. Thus the ego AVs need to consider their influence on interacting objects, balance their benefits with others, and even divide potential harms [4], [5]. Furthermore, the responsibilities to themselves and others need to be quantified and balanced.

To improve social compatibility, the first and foremost question is what factors to optimize or assign. There are currently three main factors, functional utility, regret-based utility, and communicative utility.

The functional utility includes potential harm [6], risk [7] in ethical motion planning and rewards in social compatible behavior [8]. Here responsibility depends on risk reduction when being involved in a collision outside of its reachable set [7], which does not consider the ego intention and may result in conservative regulations. The social-force-based method can also be included in this category with risk-potential as a functional utility [9]. Inverse Reinforcement Learning (IRL) is adopted to obtain the reward preference of a single driver [8], [10].

Regret-based utility focuses on the regret of certain ego actions on the reward distribution if not execute the obligation. It takes into account the aleatoric uncertainty, particularly the behavioral uncertainty of the interaction. Courtesy encourages influencing others less to prevent the ego from being too selfish [11], [12]. We propose that it can be adapted to quantify responsibilities. Further, current studies fall short of

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researching the responsibility to the ego itself to execute its intention without interference. Q-learning [13] and inverse Q Learning [14] are adopted to learn the Q function of AVs by taking certain actions and states. Yet they do not distinguish between the influences of actions on the ego and others.

Communicative utility focuses on epistemic uncertainty such as unknown parameters and needs the ego to infer the interacting object parameters from the interaction as communication [12], [15]. This is particularly valuable when it comes to adapting to changes in the environment [16]. Moreover, dealing with obstacle occlusions also requires the communicative ability to adjust to perceptual uncertainty [17]. Conversely, the CV driver's probability of observing AV is also taken into account for social compatibility [18].

In this study, we adopt the regret-based model to address the quantification of responsibilities. Our definition of responsibility employs obligation and accountability, consistent with the ethical recommendations of the EU [19]. The key insight is that a socially adept driver not only prioritizes social preferences of functional utility but also acknowledges and takes accountability for their actions and how they may impact the potential outcomes for other vehicles sharing the road. This also helps to reduce behavioral uncertainty of interactions by obligation-oriented strategies and therefore unfreeze the AVs in a highly interactive environment.

1) We build a responsibility model for autonomous driving that defines obligation and accountability in consistency with the ethical recommendations of the EU. The responsibilities are incorporated into a socially compatible driver behavior model. The responsibility towards the AV itself to execute its intention is proposed in this study as self-respect responsibility. Self-respect and courtesy responsibilities are intertwined to create a plane of reciprocal responsibility, which measures the extent of interaction intensity and tendency.
2) We introduce a hierarchical multi-agent IRL framework to calibrate the social compatibility preference of the ego driver and the functional preference of interacting vehicles from a human-driving dataset and verify the proposed model. The calibration results indicate the different contributions of modeling factors to drivers in different scenarios.
3) To further deal with the influence of interaction situations on social compatibility preference, we propose an environmental niche model as part of the weight of the responsibilities. The niche model demonstrates notable advancements in both accuracy and certainty when utilized in case analysis.

II. Related Work

A. Ethical Motion Planning

Ethical considerations are incorporated into the motion planning of autonomous vehicles to solve the fundamental problem of distributing well-being or harm [5]. Therefore responsibility is central to ethical motion planning. Two ethical frames of deontology, a rule-based ethical framework [20], [21], and consequentialism, a cost-based ethical framework [6], [7], [22], both contribute structured guidelines and formalized responsibility for vehicle behavior [23].

Nevertheless, there remains a notable gap in the literature regarding negative responsibilities in interactions based on conflicting personal interests rather than shared risks. To fill this gap, our study introduces a novel reciprocal-responsibility-based ethical driver behavior model. we present a unique approach to quantifying responsibilities towards interacting road users and integrate this framework into our comprehensive driver behavior model. Furthermore, this model delves into how situational factors can influence drivers' inclinations toward their duties in navigating the balance between conflicting responsibilities.

B. Social Interaction Model

The social interaction model includes learning-based and knowledge-based models. Knowledge-based methods for interactive agent modeling include utility-based, probability-based, field-based, and cognitive-based methods [24]. **Utility** is the most concise, resilient, and transferable representation of tasks [25].

In the field of social compatibility, the functional utility of egoism and altruism are examined. A reward function is designed based on Social Value Orientation (SVO) considering altruism and egoism functional reward [8]. In addition, a linear combination of expectation and variance of cost is also utilized for risk-aware AV [26], [27] However, the reciprocal influence was not considered here.

Based on SVO, altruism is also distinguished into cooperation and sympathy according to whether the action is reciprocal and benefits the ego vehicle [28]. The concept of reciprocal utility is adopted here to describe the influences of actions on the uncertainty of interacting agents without formal agreements [29].

The uncertainty includes that of the states and that of the parameters. We focus on the former here and assume that the ego is certain of the parameters of its interacting partners. The uncertainty of the situation gives rise to two distinct forms of regret: one stems from the potential benefits of the CV object of interaction, which is courtesy, and another is driven by the conservative tendencies of AV. The latter is proposed here to be the self-respect responsibility.

1) Courtesy: When conveying respect and showing consideration, the concept of courtesy is examined within the realm of socially compatible driving. The discrepancy in the cost of the CV attributed to AV's actions is identified as the courtesy term of reward. The effect is compared to alternatives of the best-case lane-changing scenarios [11]. However, the uncertainty of human reactions to the robot car is not considered. Considering this, an information-theoretic metric, KL divergence is adopted to describe the influence on object vehicle reward possibility distribution and measure the courtesy [12].

While courtesy is important in vehicular interactions, it alone does not account for the significance of the self-respect utility of the ego vehicle, which plays a key role in both minimizing uncertainty and facilitating predictability and trust

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in communication. This concept derives from a concession cost in persuasion.

2) Concession in Persuasion: To eliminate reliance on the learned reward function, an algorithm rooted in negotiation with persuasion and concession emerges as a strategic optimization solution. Its optimization target is added with the difference between the predicted and intended trajectory of the human driver to reflect the human driver's intention [30]. A concession cost of the weighted quadratic loss function for trajectory states is designed to describe the tendency to cling to the original planning trajectory instead of being influenced by others' actions [31]. The analysis explores how the ego vehicle's inherent conservatism is exploited by its interacting partner. Conversely, the ego also necessitates a sense of self-respect utility to maintain its focus on its path amidst the unpredictability of its interactions with others. In addressing the uncertainty inherent in the current state of affairs, a regret-based model of self-respect becomes essential. In this paper, we propose the reciprocal utilities of interaction namely self-respect utility and courtesy utility alongside functional rewards.

C. Multi-Agent Inverse Reinforcement Learning

Traffic conflicts encompass a diverse range of participants, such as both CVs and AVs, leading to complex and evolving interactions. These interactions not only introduce a higher level of complexity but also shape individual uncertainties. This poses new challenges for driver model calibrations.

Calibrations of multi-agent interactions are typically approached as the problem of inverse optimal control (IOC). The goal of IOC is to determine the most optimal cost functions that can accurately describe the trajectories of the agents [32]. The related methods can be divided into three groups, the bi-level optimization problem ([33], AZHW [34], and MSM [35]), the KKT residues minimization [36], [37], [38], and the IRL method. Although KKT can recover player objectives even from local—rather than only global—equilibrium demonstrations, it also assumes optimality while the IRL method is robust to noise. This problem of strong assumption could be mitigated with an error term in the pay-off model for bi-level optimization over likelihood [35]. However, the calibration of the proposed hierarchical behavior model is enhanced by reducing reliance on assumptions regarding the optimality of demonstrations across all parameters. In this regard, the inherent flexibility of IRL in adapting to uncertain and dynamic environments offers a more practical and robust solution.

For single-agent inverse reinforcement learning, a method of game decision-making is proposed in the form of variational inference based on maximum conditional entropy [39]. The maximum entropy approach of inverse reinforcement learning is studied under conditions of agents' ability to change the environment without dynamic models [40].

In the field of driver model calibration, reward function stability can be improved through attention mechanisms that consider scene changes [41]. To effectively employ the maximum entropy Inverse Reinforcement Learning (IRL) framework to uncover the human driver's reward, a structural assumption regarding human driving behavior is proposed. This assumption centers on identifying discreet latent states encompassing the generation, evaluation, and selection of driving trajectories. [10].

Considering the context of Multi-Agent Inverse Reinforcement Learning (MA-IRL), the calibration methods for individual agents were incapable of describing the interactions between vehicles.

Multi-Agent Inverse Reinforcement Learning(MA-IRL) is therefore developed to consider the interaction among traffic participants in model calibrations [42]. Parameter-Sharing Generative Adversarial Imitation Learning (PS-GAIL) is adopted to tackle multi-agent driver modeling with non-linearity and large state and action space [43]. The adversarial training framework utilized by them can lead to sensitivity towards hyperparameter selection and minor implementation nuances. To mitigate this issue, the concept of inverse soft-Q Learning has been introduced as a method to directly identify the Q function of the driver [14]. However, to the best of our knowledge, the causality of social responsibility has not been clarified with learning-based methods.

To calibrate the social preference of drivers, SVO is adapted with collision avoidance as a group value. This prosocial preference is calibrated with inverse reinforcement learning and compared with the online estimated SVO on Wasserstein distance of prediction error. [44]

In order to further deal with the behavioral uncertainty in interactions, risk-aware MA-IRL are developed to adopt entropy to measure and minimize uncertain. Risk-awareness, like regret-based utility, describes the tendencies to reduce behavioral uncertainty, which include reducing the ego's policy uncertainty [45] or its reward uncertainty [46] with the uncertainty of others' policies. However, the multi-agent IRL here optimizes the entropic cost of the ego instead of calibrating it for interacting partners. In addition, the block coordinate descent learning method relies on the strong assumption that the parameters of all interacting parties are true common knowledge.

Current methods have neglected the fine-tuning of social responsibility preferences in the presence of conflicts of interest across various belief systems and situational contexts. In addressing this challenge, we adopt KL divergence to define responsibilities and assess behavioral uncertainties. Leveraging quantal response equilibrium theory, our study proposes a unique hierarchical approach to MA-IRL. This method aims to accurately calibrate individual responsibilities and address the reliance of high-level responsibility features on low-level functional feature preferences. Additionally, our proposed approach allows for the calibration of a human driver's first-order beliefs about others and considers the influence of interaction conditions.

III. Problem Formulation

To model human drivers' socially compatible behavior, we focus on a two-vehicle lane-change interaction in traffic. We aim to model the mandatory and discretionary lane change behavior of the ego vehicle (denoted by $()_E$) in the interaction

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of the object vehicle (denoted by $()_O$) in the target lane. The driver of the ego vehicle observes state $\mathbf{x}_t \in \mathcal{X}$ of the interacting vehicles. The observation is adopted to predict the probabilities of actions $\mathbf{u}_t \in \mathcal{U}$ of the interacting vehicles. With the assumption of a discrete-time setup and a finite time horizon T, a trajectory $\zeta = [\mathbf{x}_1, \mathbf{u}_1, \mathbf{x}_2, \mathbf{u}_2 \ldots, \mathbf{x}_T, \mathbf{u}_T]$ is organized within the decision horizon.

A sample-based planner setting is adopted for each vehicle [10]. Both vehicles are presumed to be noisy optimizers, seeking to optimize either the cumulative reward R_E or R_O over the sampled trajectories. We further assume that they perform the Stackelberg game, where the ego vehicle is the leader and the object follower [11]. In light of these assumptions and within a rolling horizon framework, from the perspective of the ego vehicle, the object is perceived as a naive follower seeking to maximize the function:

$$\zeta_O^* = \arg\max_{\zeta_O} R_O\left(\mathbf{x}^0, \zeta_O, \zeta_E\right) \tag{1}$$

Here, $\mathbf{x}^0$ represents the initial state, ζ_O denotes the specified trajectory of the object, and ζ_E signifies the given ego trajectory. The probability of the interacting object vehicle choosing its trajectory, according to the Boltzmann noisily-rational model, is exponential to the cumulative reward of the trajectory [47].

$$\zeta_O\left(\mathbf{x}^0, \zeta_E\right) \sim P\left(\zeta_O \mid \mathbf{x}^0, \zeta_E\right) \propto e^{\beta R_O\left(\mathbf{x}^0, \zeta_E, \zeta_O\right)} \tag{2}$$

where β regulates the irrationality of the driver. To simplify the problem without any loss of generality, we assume $\beta = 1$ and exclude it for conciseness.

Taking into account Equation (2), the autonomous vehicle optimizes its trajectories according to the following equation:

$$\begin{aligned}\zeta_E^* &= \arg\max_{\zeta_E} R_E\left(\mathbf{x}^0, \zeta_E, \zeta_O \sim P\left(\zeta_O \mid \mathbf{x}^0, \zeta_E\right)\right)\\ &= \arg\max_{\zeta_E} R_E\left(\mathbf{x}^0, \zeta_E\right)\end{aligned} \tag{3}$$

To model the social compatibility of human drivers, we introduce the prediction and planning method in Section IV and the calibration for it with the human driving dataset in Section V.

IV. Prediction and Planning With Social Compatibility Utilities

A. Social Compatibility Utilities

While the object vehicle bases its decisions on functional rewards, the ego vehicle is assumed to determine its behavioral distribution $P\left(\zeta_E \mid \mathbf{x}^0\right)$ according to social compatibility utilities.

$$P\left(\zeta_E \mid \mathbf{x}^0\right) = \frac{e^{R_E\left(\mathbf{x}^0, \zeta_E\right)}}{\sum_{\tilde{\zeta}^i \in \mathbf{D}_E} e^{R_E\left(\mathbf{x}^0, \tilde{\zeta}^i\right)}} \tag{4}$$

where $R_E\left(\mathbf{x}^0, \zeta_E\right)$ denotes the ego vehicle's total social compatibility reward, $\tilde{\zeta}^i$ is a generated trajectory of the object that has the same initial state as ζ_E, $\mathbf{D}_E$ is the generated trajectory set.

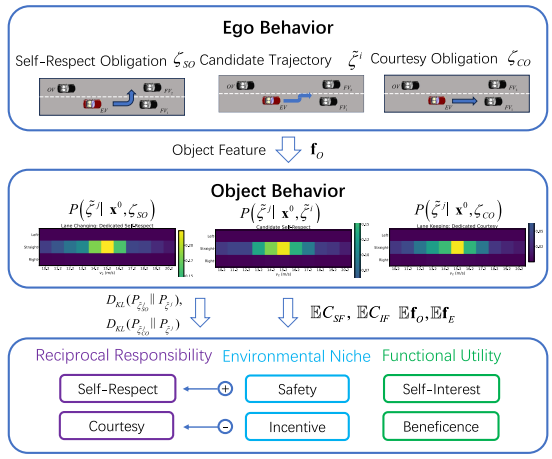

Fig. 1. Illustration of the relationship among social compatibility utilities.

R_E is determined by the social-compatible features and weights $\boldsymbol{\theta}_{sc}, \mathbf{F}_{\zeta_E}$.

$$R_E\left(\mathbf{x}^0, \zeta_E\right) = \boldsymbol{\theta}_{sc}{}^T \sum_{t=0}^{N-1} \mathbf{F}_{\zeta_E}\left(\mathbf{x}^t, \zeta_E^t\right) \tag{5}$$

where $\mathbf{F}_{\zeta_E} \in R^4$ denotes four higher-level social-compatible utilities. $\boldsymbol{\theta}_{sc} \in R^4$ denotes their weights.

$$\begin{aligned}\mathbf{F}_{\zeta_E}\left(\mathbf{x}^t, \zeta_E^t\right) = \Big[&R_{E_{\mathrm{SI}}}\left(\mathbf{x}^0, \zeta_E\right), R_{E_{\mathrm{SR}}}\left(\mathbf{x}^0, \zeta_E\right),\\ &R_{E_{\mathrm{court}}}\left(\mathbf{x}^0, \zeta_E\right), R_{E_{\mathrm{benef}}}\left(\mathbf{x}^0, \zeta_E\right)\Big]\end{aligned} \tag{6}$$

where the $R_{E_{\mathrm{SI}}}$, $R_{E_{\mathrm{SR}}}$, $R_{E_{\mathrm{court}}}$, and $R_{E_{\mathrm{benef}}}$ are respectively the ego vehicle's self-interest utility, self-respect utility, courtesy utility, and beneficence utility.

They can be categorized into two main sections: egoism and altruism. Both of these categories comprise two classes of utilities known as functional and reciprocal responsibility utility, as illustrated in Figure 1.

Figure 1 shows the relationship among the social compatibility utilities. $P\left(\zeta_O \mid \mathbf{x}^0, \zeta_E\right)$ is utilized to represent the uncertainty of the interacting object vehicle behavior. This probability distribution allows for the computation of the functional reward that the object may receive, since the functional utilities of both sides have the same distribution in the framework of the Stackelberg game.

Functional utilities refer to the anticipated benefits that result from the interaction of vehicles, which are determined by their characteristics in terms of safety, efficiency, comfort, and energy conservation.

Reciprocal responsibilities utilities emphasize the impact of the trajectory ζ_E selected by the ego vehicle on the object behavior uncertainty, as opposed to when the ego's obligations are in effect. In a lane-changing context, the self-respect obligation normatively demands the ego vehicle to change lanes at constant speed following the preceding vehicle. Meanwhile, the courtesy obligation requires the ego to cruise on the original lane.

This impact offers opportunities to stabilize the interaction by certain ego actions, decrease the joint behavioral uncertainty, and carve the ego's environmental niches. Specifically,

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there are two factors in the environmental niches, the safety factor and the incentive factor. Improving safety measures on the road can positively impact drivers' confidence in changing lanes, leading to a greater sense of self-respect. Meanwhile, the presence of incentives may diminish drivers' inherent motivation to act courteously towards others.

B. Functional Utility

Functional utility describes the drivers' expected rewards whose uncertainty is mainly caused by others' uncertain behavior choices.

The functional utility of an ego agent can refer to Eq. (3) and be calculated considering the behavior distribution $P\left(\zeta_O \mid \mathbf{x}^0, \zeta_E\right)$ of the interacting object vehicle

$$P\left(\zeta_O \mid \mathbf{x}^0, \zeta_E\right) = \frac{e^{R_O\left(\mathbf{x}^0, \zeta_E, \zeta_O\right)}}{\sum_{\tilde{\zeta}^j \in \mathbf{D}_O} e^{R_O\left(\mathbf{x}^0, \tilde{\zeta}_E, \tilde{\zeta}^j\right)}} \tag{7}$$

where $R_O\left(\mathbf{x}^0, \zeta_E, \zeta_O\right)$ denotes the object vehicle's functional rewards. In this context, we exclusively focus on the functional rewards of the object, while the social compatibility aspects are evaluated for the ego vehicle.

The object vehicle's functional reward in the horizon of t is defined with the current state $\mathbf{x}^0$, the candidate trajectory pair (ζ_E, ζ_O), and the candidate action pair (u_E^t, u_O^t) at time t

$$R_O\left(\mathbf{x}^0, \zeta_E, \zeta_O\right) = \boldsymbol{\theta_2}^T \sum_{t=0}^{N-1} \mathbf{f}_O\left(\mathbf{x}^t, \zeta_E^t, \zeta_O^t\right) \tag{8}$$

where $\mathbf{f}_O \in R^8$ denotes eight lower-level functional rewards. A more detailed description can be found in VI-B. $\boldsymbol{\theta_2} \in R^8$ denotes their weights, which is learned via MA-IRL.

Thus the functional utility of the ego vehicle and the other can be calculated, respectively.

1) Self-Interest: The functional utility of the ego vehicle is the self-interest utility.

$$R_{E_{\mathrm{SI}}}\left(\mathbf{x}^0, \zeta_E\right) = \mathbb{E}_{\tilde{\zeta}^j \sim P\left(\zeta_O \mid \mathbf{x}^0, \zeta_E\right)}\left[R_{E_{\mathrm{func}}}\left(\mathbf{x}^0, \zeta_E, \tilde{\zeta}^j\right)\right] \tag{9}$$

where

$$R_{E_{\mathrm{func}}}\left(\mathbf{x}^0, \zeta_E, \zeta_O\right) = \boldsymbol{\theta_1}^T \sum_{t=0}^{N-1} \mathbf{f}_E\left(\mathbf{x}^t, \zeta_E^t, \zeta_O^t\right) \tag{10}$$

here $\mathbf{f}_E \in R^8$ denotes eight lower-level functional rewards if this is a discretionary lane-change while $\mathbf{f}_E \in R^9$ if this is a mandatory lane-change. Meanwhile $\boldsymbol{\theta_1} \in R^8$ or $\boldsymbol{\theta_1} \in R^9$ denotes their weights. Detailed analysis over the functional features is elaborated in section VI-B. ζ_E is the chosen trajectory of the ego vehicle.

2) Beneficence: The beneficence utility describes how altruistic the ego vehicle is for the expected functional utility of the other interacting vehicle.

$$R_{E_{\mathrm{benef}}}\left(\mathbf{x}^0, \zeta_E\right) = \mathbb{E}_{\tilde{\zeta}^j \sim P\left(\zeta_O \mid \mathbf{x}^0, \zeta_E\right)}\left[R_O\left(\mathbf{x}^0, \zeta_E, \tilde{\zeta}^j\right)\right] \tag{11}$$

here the $\tilde{\zeta}^j$ is a choice of the object vehicle trajectory.

C. Reciprocal Responsibility Utility

Responsibility was considered with the reachable set for the behavioral uncertainty [7]. Nonetheless, vehicles can possess conflicting intentions that are not necessarily addressed by a common safety concern. As a result, it is essential to take into account intentions and moral assertions when assigning responsibility. Most analyses share that responsibility should based on a causal connection between the person and the outcome of actions (causality), the subject should know the consequences (intentionality), and they should be able to choose to act freely (reachability) [48]. We adopt the KL divergence of influenced possibility distribution on objects to infer causality. We assume that the ego is certain about the objects' preference which indicates intentionality. For reachability, we take the reachable best cases as the baseline obligation for responsibility computing.

Reciprocal responsibility is defined as the mutual commitment among members of a community to fulfill their shared obligations without interference or hindrance from others [49]. In this study, we adopt it to encompass two key elements: self-respect and courtesy, which describe the influences of an ego trajectory on both of the interacting vehicles. Therefore, a reciprocity responsibility plane can be formed and describe the interaction intensity as well as the tendency.

1) Self-Respect: Self-respect is defined as one's commitment to its self-respect obligation, unwavering in the face of outside influences that may tempt one to stray from this trajectory.

We propose to approximate the change made to the reward distribution of the ego vehicle with the transport of the object trajectory's possibility distribution. This is because the ego's functional reward depends on the trajectory choice of the object. The transport is computed with the baseline of a reachable self-respect obligation. For a lane-changing vehicle, naturally, the self-respect obligation trajectory $\zeta_{E_{SO}}$ is set to be changing the lane with a target speed of v_{E-SO} following its preceding vehicle in the target lane.

$$v_{E-SO} = \min\{v_{E_0}, \max\{v_{E_{min}}, v_{FT}\}\} \tag{12}$$

where the v_{E_0} is the original velocity of the ego at the start, $v_{E_{min}}$ is the minimum of the target longitudinal velocity set, v_{FT} is the velocity of the preceding vehicle in the target lane. The corresponding probability distribution with v_{E-SO} as target longitudinal velocity is denoted $P\left(\zeta_O \mid \mathbf{x}^0, \zeta_{E_{SO}}\right)$, $P_{\tilde{\zeta}_{SO}}$ for short.

The Kullback-Leibler (KL) divergence [50] is adopted to capture the difference between the changes. Especially, to cultivate a mode-covering behavior of KL divergence [51] and cover the mode of the self-respect obligation , we adopt the following expression for self-respect reward.

$$\begin{aligned} R_{E_{SR}}\left(\mathbf{x}^0, \zeta_E\right) &= e^{\left[\mathbb{E}_{\tilde{\zeta}^j}\left[C_{SF-\tilde{\zeta}^j}\right] - D_{KL}\left[P\left(\zeta_O \mid \mathbf{x}^0, \zeta_{E_{SO}}\right) \| P_{\tilde{\zeta}^j}\right]\right]} \\ &= e^{\sum_{j=1}^{M} P_{\tilde{\zeta}^j} C_{SF-\tilde{\zeta}^j} - \sum_{j=1}^{M} P_{\tilde{\zeta}_{SO}^j} \log \frac{P_{\tilde{\zeta}_{SO}^j}}{P_{\tilde{\zeta}^j}}}. \end{aligned} \tag{13}$$

Here the exponent has two components. On one hand, the KL divergence infers accountability between a candidate

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trajectory ζ_E and obligation $\zeta_{E_{SO}}$. On the other hand, the $C_{SF-\tilde{\zeta}^j}$ denotes the safety factor as the consequence of the trajectory $\tilde{\zeta}^j$. M denotes the total number of generated ego trajectories. It describes the value of the trajectory terminal state, which will be elaborated in IV-D.1 and IV-D.2.

2) Courtesy: Courtesy, altruistic reciprocal responsibility, highlights the importance of considering others' plans and minimizing disruptions to their intentions [12]. It is also defined based on an obligation. In the case of lane changing, the courtesy obligation of the ego is chosen as cruising at a target speed of $v_{E_{CO}}$ and not changing lanes, which has the least impact on the object.

$$v_{E_{CO}} = \min\{v_{E_0}, \max\{v_{E_{min}}, v_{FC}\}\} \tag{14}$$

where v_{FC} is the velocity of the preceding vehicle in the current lane. The corresponding probability distribution with v_{E-CO} as target longitudinal velocity is denoted $P\left(\zeta_O \mid \mathbf{x}^0, \zeta_{E_{CO}}\right)$, $P_{\tilde{\zeta}_{CO}}$ for short.

$$\begin{aligned} R_{E_{\text{court}}}\left(\mathbf{x}^0, \zeta_E\right) &= e^{\left[\mathbb{E}_{\tilde{\zeta}^j}\left[C_{IF-\tilde{\zeta}^j}\right] - D_{KL}\left[P\left(\zeta_O \mid \mathbf{x}^0, \zeta_{E_{CO}}\right) \| P_{\tilde{\zeta}^j}\right]\right]} \\ &= e^{\sum_{j=1}^{M} P_{\tilde{\zeta}^j} C_{IF-\tilde{\zeta}^j} - \sum_{j=1}^{M} P_{\tilde{\zeta}_{CO}^j} \log \frac{P_{\tilde{\zeta}_{CO}^j}}{P_{\tilde{\zeta}^j}}}. \end{aligned} \tag{15}$$

where $C_{IF-\tilde{\zeta}^j}$ denotes the Incentive Factor as the consequence of the trajectory $\tilde{\zeta}^j$, which will be elaborated in IV-D.3.

3) Self-Respect-Courtesy (SR-C) Plane: A Self-Respect-Courtesy (SR-C) plane is formed with self-respect and courtesy responsibility rewards as shown by Fig. 2. It can describe the interaction intensity and tendency based on responsibilities and accountabilities.

The outer functions of the self-respect and courtesy rewards are both exponential functions and therefore positive according to (13) and (15). In order to consider solely the reciprocal responsibility of trajectories, we set the trajectory terminal conditions as zero and obtain the remaining self-respect and courtesy responsibility rewards. KL divergence is nonnegative and it becomes zero if and only if $P_{\tilde{\zeta}^j} = P_{\tilde{\zeta}_{CO}^j}$ here. Therefore the responsibility rewards range between $(0, 1]$ and they become one if and only if they have the same influence on the object as the corresponding obligation. The remaining two responsibility utilities form the SR-C plane as Fig. 2.

Therefore, the interaction responsibility can be defined in a plane.

Figure 2 illustrates the extremal values of the two responsibility rewards, forming a rectangular envelope that delineates the reachable trajectory responsibility set. Accountability, defined as the distance between the candidate case and the obligation case, quantifies the extent to which the ego can be held responsible for fulfilling its duties [19].

Conversely, comparing the candidate case with the worst-case scenario reveals the actual degree of responsibility achieved in practice.

Combined, the distance between the best and worst-case scenarios reflects the ego's capacity to influence the object and signifies the full responsibility of the vehicle. This underscores the principle that with great power comes great responsibility.

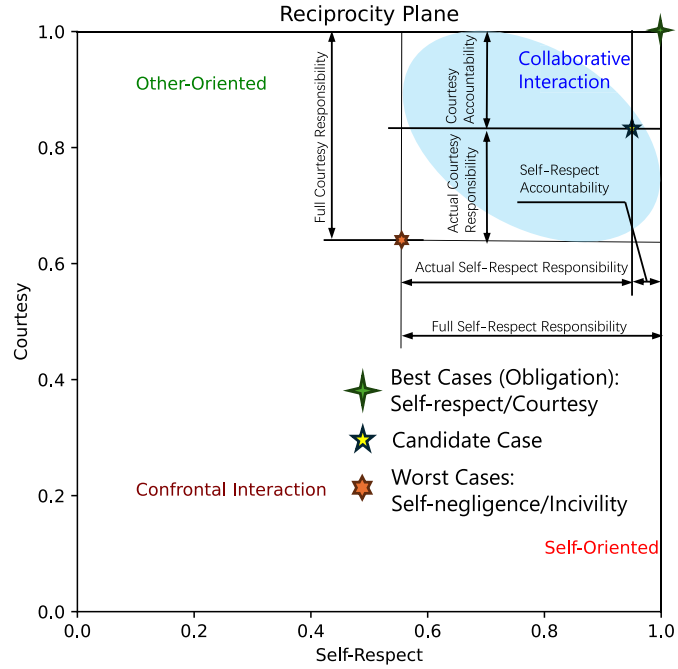

Fig. 2. The Self-Respect-Courtesy (SR-C) plane with the accountability, actual responsibility, and full responsibility of the ego to both interacting vehicles.

This also indicates the behavioral uncertainty that the ego needs to navigate in the interaction.

One of the diagonal lines of the SR-C plane shows interaction intensity. The point $(0, 0)$ represents a confrontational interaction with the highest level of accountabilities, whereas the point $(1, 1)$ signifies a collaborative interaction with minimal accountabilities and conflicts. The diagonal line between these two points indicates the tendency towards different types of interactions. The coordinate $(1, 0)$ is the most self-oriented while $(0, 1)$ most other-oriented.

D. Environmental Niches

We utilize the ecological concept of a niche to delineate the environment in which resources of safety and incentive influence the interaction between the ego and the object vehicle, which are coexisting consumers. Environmental niches include the impact niche and the requirement niche of the ego to describe its interaction conditions [52]. While conditions are considered in autonomous driving decision-making or prediction [53], the niches we propose here describe the anticipated conditions as a result of certain ego and object trajectories. A similar idea was modeled as time-variant constraints [54]. In this case, we optimize them as parts of the rewards. The ego strategically carves out its place in order to shoulder greater responsibilities and foster stable interactions. One illustrative instance of this is when the ego accelerates longitudinally in preparation for a lane change, to ensure safety and minimize behavior uncertainty.

Here we focus on the safety and incentive factors as niches for lane-changing decisions. The safety factors are defined with an adapted probabilistic condition model from [53].

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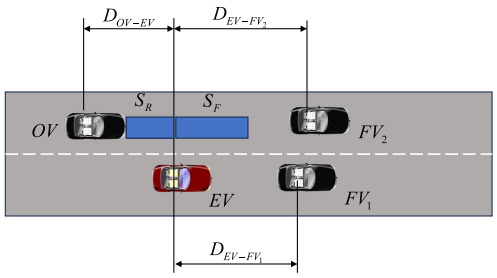

Fig. 3. Layout of the lane change process for the probabilistic model of niches.

Intuitively, the ego may prioritize safety factors over upholding a sense of self-respect in a challenging situation like a mandatory lane change. The safety factor C_{SF} combines two parts, the longitudinal car following safety factor C_{CF} and the lateral lane-changing safety factor C_{LC}.

$$C_{SF} = \min\{C_{CF}, C_{LC}\} \tag{16}$$

1) Car Following Safety Factor: The expected condition of lane change gap $\mathbb{E}_{\tilde{\zeta}j}\left[C_{\tilde{\zeta}j}\right]$ in Eq. (13) depends on the distribution of object trajectories. It captures the influence of the gap condition on the self-respect of the ego vehicle.

Safety condition consists of two parts, the safety preceding gap criterion, and the safety rear gap criterion. Each of the criteria is designed with relative distance and Gaussian distribution assumption.

$$\begin{aligned} C_{CF} &= P\left(\text{ Safe }_{\mathrm{F}} \mid D_{\mathrm{OV-EV}}\right) P\left(\text{ Safe }_{\mathrm{R}} \mid D_{\mathrm{EV-FV}}\right) \\ P\left(\text{ Safe }_{\mathrm{F}} \mid D_{\mathrm{OV-EV}}\right) &= \mathcal{N}\left(D_{\mathrm{OV-EV}} \mid \mu_1, \sigma_1^2\right), \\ P\left(\text{ Safe }_{\mathrm{R}} \mid D_{\mathrm{EV-FV}}\right) &= \mathcal{N}\left(D_{\mathrm{EV-FV}} \mid \mu_2, \sigma_2^2\right), \end{aligned} \tag{17}$$

where $D_{\mathrm{OV-EV}}$ and $D_{\mathrm{EV-FV}}$ denotes the observed distance between related vehicles as in Figure 3. Here FV are chosen from the nearest preceding vehicle in both current and target lanes.

We adopt the preceding gap instead of the distance from the object vehicle to the front [53] to avoid that a large $D_{\mathrm{OV-EV}}$ may hide a risk situation of small gap $D_{\mathrm{EV-FV}}$. The Gaussian parameters are designed as $\mu_1 = 10, \mu_2 = 15, \sigma_1^2 = 5, \sigma_2^2 = 5$.

2) Lane-Changing Safety Factor: The lane-changing safety factor adapts from a passing/yielding inference probabilistic condition [53]. The assumption is simplified from constant acceleration to constant velocity for the evaluation of future state value instead of action advantage.

In particular, we calculate the motion state boundaries for passing and yielding, E_{pass} and E_{yield} (e.g. minimal lateral velocity for passing and maximal lateral velocity for yielding), and the behavioral status is then observed by comparing the boundaries with the vehicle's current motion state E:

$$\begin{aligned} D_{\text{pass}} &= \left(E - E_{\text{yield}}\right) / \left(E_{\text{pass}} - E_{\text{yield}}\right), \\ C_{LC} &= P\left(B = \text{ pass } \mid D_{\text{pass}}\right) = \mathcal{N}\left(D_{\text{pass}} \mid \mu_1, \sigma_1^2\right) \end{aligned} \tag{18}$$

the Gaussian parameters are set as $\mu_1 = 1, \sigma_1^2 = 0.15$. E_{yield} is set as -0.15m/s.

The E_{pass} refers to the minimal lateral velocity for the ego vehicle (EV) to cross the lane line before the terminal. We apply the assumptions of constant velocity to predict vehicles' positions at the terminal.

$$E_{\text{pass}} = \frac{D_y}{t_{terminal}} \tag{19}$$

where the D_y denotes the remaining lateral distance to the target lane center line of the ego vehicle.

$$t_{terminal} = \min\{t_{lane-tmn}, t_{dec}, T_0, t_{EV-OV}, t_{EV-FV_2}\} \tag{20}$$

where the terminal $t_{terminal}$ denotes the time when the object vehicle (OV) reaches the lane terminal at $t_{lane-tmn}$ or EV's velocity decreases to 0 at t_{dec}. If the EV is too far from the lane terminal, it is set as $T_0 = 4s$ from the current time. For passing, the ego needs minimal time for EV to lead OV before t_{EV-OV} and keep a distance to FV_2 before t_{EV-FV_2} under the assumption of constant velocity.

3) Incentive Factor: The incentive factor has a negative influence on courtesy. Therefore we adopt the following model of incentive factor C_{IF} inspired by the IDM [55].

$$\begin{aligned} C_{IF} &= \min\{1, \frac{v_{LK}}{v_{LC}}\} \\ v_{LK} &= \min\{v_{\zeta_E}, v_{FV_1}\} \\ v_{LC} &= \min\{v_0, v_{FV_2}\} \end{aligned} \tag{21}$$

where v_{LC} represents the anticipated speed after changing lanes, while v_{LK} signifies the projected speed if no lane change occurs. v_{ζ_E} is the terminal ego velocity of a trajectory ζ_E, while v_0 signifies a desired velocity, which is 20m/s here. The upper limit of the variable C_{IF} is established at 1 to maintain compatibility with discretionary lane changes.

V. Hierarchical MA-IRL

A. Social Compatibility Multi-Agent Inverse Reinforcement Learning

A hierarchical social compatibility inverse reinforcement learning (SC-MIRL) is constructed, which integrates the calibration and sampling of the interacting object vehicle. The planning and corresponding calibration algorithm is shown in Fig. 4.

1) Hierarchical Multi-Agent Inverse Reinforcement Learning: The hierarchical parameters of the social-compatible rewards system are calibrated with SC-MIRL. The preference weights of the interacting vehicles, lower level functional weights $\boldsymbol{\theta_1}, \boldsymbol{\theta_2}$, and social compatibility weight $\boldsymbol{\theta_{sc}}$ are estimated with functional features, $\mathbf{f}_E$ and $\mathbf{f}_O$.

2) Trajectories Sampling: The functional features are subtracted based on a pairwise trajectories sampling scheme for constructing a joint behavior space and forming a pairwise interactive environment.

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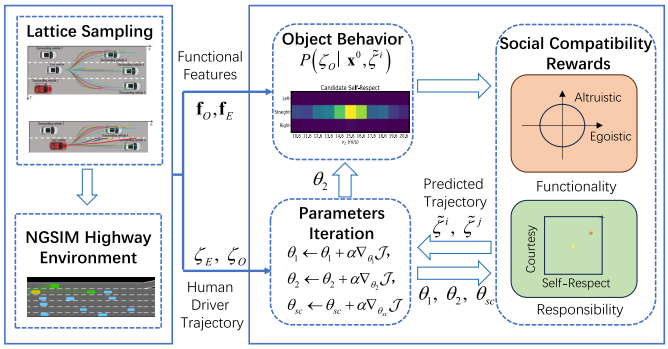

Fig. 4. Illustration of the social compatibility calibration framework.

B. Hierarchical Multi-Agent Inverse Reinforcement Learning

The single-agent Inverse Reinforcement Learning (IRL) method can calibrate the driver model, but it relies on the assumption that the ego vehicle accurately knows the exact actions of other interacting objects. Besides it is not ideal for calibrating hierarchical weights of the social compatibility rewards. As a solution, we introduce a novel algorithm called Social Compatibility Multi-Agent Inverse Reinforcement Learning (SC-MIRL). This method operates based on certain first-order beliefs, rather than presuming full knowledge of other agents' actions.

The goal of the multi-agent inverse reinforcement learning is to identify the optimal parameters, $\boldsymbol{\theta_1}$, $\boldsymbol{\theta_2}$, and $\boldsymbol{\theta_{sc}}$, as depicted in Figure 4, to effectively increase the likelihood of human driver demonstration trajectories observed in three-second sampled trajectories.

$$\max_{\boldsymbol{\theta_1},\boldsymbol{\theta_2},\boldsymbol{\theta_{sc}}} \mathcal{J}(\boldsymbol{\theta_1}, \boldsymbol{\theta_2}, \boldsymbol{\theta_{sc}}) = \max_{\boldsymbol{\theta_1},\boldsymbol{\theta_2},\boldsymbol{\theta_{sc}}} \sum_{\tilde{\zeta}^i \in \mathbf{D}_E, \tilde{\zeta}^j \in \mathbf{D}_O} \log P\left(\tilde{\zeta}^i \mid \mathbf{x}^0\right) \tag{22}$$

where $\mathbf{D}_E = \{\tilde{\zeta}^i\}_{i=1}^N$, $\mathbf{D}_O = \{\tilde{\zeta}^j\}_{j=1}^N$ is the trajectory sets of human demonstrations for the two interacting vehicles.

Substituting (22) with (5), the object function $\mathcal{J}$ can be rebuilt as:

$$\mathcal{J}(\boldsymbol{\theta_{sc}}, \boldsymbol{\theta_1}, \boldsymbol{\theta_2}) = \sum_{\tilde{\zeta}^i \in \mathbf{D}_E, \tilde{\zeta}^j \in \mathbf{D}_O} \left[\boldsymbol{\theta_{sc}}^T \mathbf{F}_{\zeta^E} - \log \sum_{i=1}^{M} e^{\boldsymbol{\theta_{sc}}^T \mathbf{F}_{\tilde{\zeta}^i}} \right] \tag{23}$$

where $\tilde{\zeta}^i$ is a trajectory that originates from the same initial state as ζ^i.

To optimize $\mathcal{J}$ with a gradient-based method, the partial derivatives of this function on three sets of weights $\boldsymbol{\theta_1}$, $\boldsymbol{\theta_2}$, and $\boldsymbol{\theta_{sc}}$ are calculated as:

$$\nabla_{\boldsymbol{\theta_{sc}}} \mathcal{J} = \sum_{\tilde{\zeta}^i \in \mathbf{D}_E, \tilde{\zeta}^j \in \mathbf{D}_O} \left[\mathbf{F}_{\zeta_E} - \mathbb{E}_{\tilde{\zeta}^i \sim P\left(\tilde{\zeta}^i\right)} \left[\mathbf{F}_{\tilde{\zeta}^i} \right] \right] \tag{24}$$

$$\nabla_{\boldsymbol{\theta_1}} \mathcal{J} = \sum_{\tilde{\zeta}^i \in \mathbf{D}_E, \tilde{\zeta}^j \in \mathbf{D}_O} \left[\boldsymbol{\theta_{sc}}^T \nabla_{\boldsymbol{\theta_1}} \mathbf{F}_{\zeta_E} - \mathbb{E}_{\tilde{\zeta}^i \sim P\left(\tilde{\zeta}^i\right)} \left[\boldsymbol{\theta_{sc}}^T \nabla_{\boldsymbol{\theta_1}} \mathbf{F}_{\tilde{\zeta}^i} \right] \right] \tag{25}$$

$$\nabla_{\boldsymbol{\theta_2}} \mathcal{J} = \sum_{\tilde{\zeta}^i \in \mathbf{D}_E, \tilde{\zeta}^j \in \mathbf{D}_O} \left[\boldsymbol{\theta_{sc}}^T \nabla_{\boldsymbol{\theta_2}} \mathbf{F}_{\zeta_E} - \mathbb{E}_{\tilde{\zeta}^i \sim P\left(\tilde{\zeta}^i\right)} \left[\boldsymbol{\theta_{sc}}^T \nabla_{\boldsymbol{\theta_2}} \mathbf{F}_{\tilde{\zeta}^i} \right] \right] \tag{26}$$

The partial derivative $\nabla_{\boldsymbol{\theta_{sc}}} \mathcal{J}$ equals the discrepancy in social compatibility rewards between human demonstration trajectories ζ_E and expectations over the generated ones $\tilde{\zeta}^i$. This is consistent with the expectation difference as the derivative in IRL.

Algorithm 1 summarizes the algorithm steps. Furthermore, the hierarchical structure of the weights and the game-theoretical model of interaction introduce a possibility of mutual interference in the convergence of their gradient descent iterations. Therefore, simultaneously updating the gradient of weights doesn't ensure convergence [56]. To address this challenge, we utilize an asynchronous iteration method as shown in Algorithm 1, drawing inspiration from the Generative Adversarial Network (GAN) framework [57]. The parameters are established at $k_1 = 5$ and $k_2 = 16$. The condition of convergence is quantified by calculating the total log-likelihood expected from the last k_2 iteration steps.

In the proposed method, we can update the ego's first-order belief of the object preference $\boldsymbol{\theta_2}$ with actions of the ego. If only the parameter $\boldsymbol{\theta_1}$ is updated, the proposed algorithm reverts to a single-agent maximum entropy Inverse Reinforcement Learning (IRL). When both parameters $\boldsymbol{\theta_1}$ and $\boldsymbol{\theta_2}$ are updated, the algorithm results in a Multi-Agent IRL (MA-IRL) without taking into account social compatibility. This approach is adopted in VII as the baseline for comparison.

C. Trajectories Sampling

Two interpolating polynomial curve planners are employed to configure the planning space for the interacting vehicles with target points. Point-based curves are particularly effective in geometrically constrained environments such as highways and ensure the dynamic constraints of the vehicles [58]. In the longitudinal direction, drivers are responsible for determining the desired speed, while decisions regarding tactical lane changes are made in the lateral direction [10].

We especially adopt a context-aware lane-changing planner for the ego vehicle in lateral directions [59]. This planner enlarges the solution space and improves the forward simulation's success rate. The resulting decision can adapt to the traffic configuration more flexibly. It is here adapted to realize complex semantic-level action of lane-changing anticipation in a simple setting of interpolating polynomial curve planning.

Fig. 5a is an example of the trajectories sampling process. The target space of two interacting vehicles in a horizon T of 3s is specified as $\Phi = \{v_{x_E}, y_E, v_{x_O}, y_O\}$. A lattice of polynomial trajectories is therefore computed with quartic and quintic polynomials [60] and the velocity and position at every instant in the horizon are determined.

Figure 5b describes the lateral target choices of a context-aware lane-changing process for the ego. The process is divided into three phases lane-keeping, lane-changing anticipation, and lane-changing. They are divided by the lane-changing anticipation lines, represented by γ_a and γ_d, as the vehicle

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Algorithm 1 Socially-Compatibility Multi-Agent Inverse Reinforcement Leaning

Inputs
system dynamics P, agents' cost features $\mathbf{f}_{\tilde{\zeta}^i}$, and the set of interaction demonstrations.
Initialization
for each agent $i \in [N]$, generate sampled trajectory set, rollout with P, compute the empirical feature under demonstrations $\mathbf{f}_{\tilde{\zeta}^i}$, and add to buffer
initialize the cost weights $\boldsymbol{\theta_1}$,$\boldsymbol{\theta_2}$ and $\boldsymbol{\theta_{sc}}$
$k = 1$
while not converged **do**
    Compute features $\mathbf{F}_{\zeta_\mathbf{E}}$, $\nabla_{\boldsymbol{\theta_1}}\mathbf{F}$, and $\nabla_{\boldsymbol{\theta_2}}\mathbf{F}$
    Compute the difference $\nabla_{\boldsymbol{\theta_{sc}}}\mathcal{J}$, $\nabla_{\boldsymbol{\theta_1}}\mathcal{J}$, and $\nabla_{\boldsymbol{\theta_2}}\mathcal{J}$

$$\boldsymbol{\theta}_{sc} \leftarrow \boldsymbol{\theta}_{sc} + \alpha \nabla_{\boldsymbol{\theta}_{sc}} \mathcal{J}$$

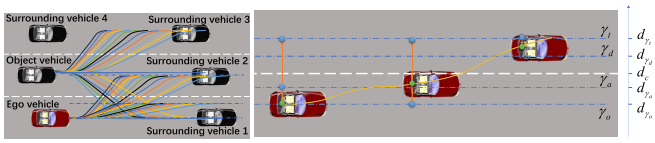
    if k is divisible by k_1 **then**

$$\boldsymbol{\theta}_1 \leftarrow \boldsymbol{\theta}_1 + \alpha \nabla_{\boldsymbol{\theta}_1} \mathcal{J}$$

    end if
    if k is divisible by k_2 **then**

$$\boldsymbol{\theta}_2 \leftarrow \boldsymbol{\theta}_2 + \alpha \nabla_{\boldsymbol{\theta}_2} \mathcal{J}$$

    end if
    $k \leftarrow k + 1$
end while
return cost parameters $\boldsymbol{\theta_1}$,$\boldsymbol{\theta_2}$ and $\boldsymbol{\theta_{sc}}$



(a) (b)

Fig. 5. The lattice planner for two lane-change interaction vehicles and context-aware lane-changing process for the ego. (a) The longitudinal and lateral lane changing planner for the trajectory sampling process. (b) The lateral target positions of a context-aware planner during a lane-changing process. The yellow line is the actual trajectory. The green dots and the yellow ones are the vehicle lane-changing lateral target points at each position. Green dots are the current lateral position.

moves across the lane line. d_{γ_a}, d_{γ_d} is their lateral positions.

$$d_{\gamma_a} = d_c - \frac{W_{ego}}{2} \tag{27}$$

$$d_{\gamma_d} = d_c + \frac{W_{ego}}{2} \tag{28}$$

where d_c is the lateral position of the lane line, the W_{ego} is the width of the ego vehicle.

At the beginning and lane-changing anticipation of the lane-changing process, the ego vehicles consider lane-keeping at the current position, lane-changing to the target lane center line γ_t, and lane-changing anticipation at γ_a. After crossing the lane line, it only considers lane keeping at γ_a or its current position.

The sampling method allows for the precise enumeration of interacting trajectories within a constrained time frame, thus making it well-suited for the proposed small-scale model that operates with only a handful of parameters. Although this simplified model may not fully capture the intricacies of trajectory representation, it serves as a solid proof-of-concept for responsible design.

D. Environment Model

Forward simulation is leveraged to generate features for interacting vehicles. It is an interactive NGSIM environment [10], which is systematically adapted for pairwise interaction simulation here to sample trajectories and construct a joint behavior space.

In this setting, nearby vehicles will follow their predetermined paths and carefully monitor the distance between themselves and the vehicles ahead. Should the gap between the ego and its follower fall below the optimal distance as stipulated by the Intelligent Driver Model (IDM), the following vehicle will adjust its course by the IDM protocol, veering away from its intended path.

The observation of the four vehicles in the lane-changing scene is also recorded for the algorithm to compute environmental niches of the safety and incentive factors.

VI. Experimental Validation

A. Naturalistic Human Driving Dataset

To validate the proposed SC-MIRL for driver model calibration, we adopt the Next Generation Simulation (NGSIM) dataset [61] with a segment of data on US Highway 101. This study area spans a distance of approximately 640 meters and is comprised of five mainline lanes with a width of 3.66 meters each. Additionally, an auxiliary lane is situated between the on-ramp and the off-ramp, as illustrated in Fig. 6a. The dataset comprises 45 minutes of data captured from 7:50 a.m. to 8:35 a.m., segmented into three distinct 15-minute periods.

The road layout and traffic are reconstructed within the environment model, as illustrated in Fig. 6a. Each vehicle's position within the area is meticulously captured at a frequency of 10 frames per second, yielding comprehensive and precise trajectory information from the initial point within the area to its conclusion. The trajectories are therefore reconstructed for scene extraction as in Fig. 6b.

Vehicle lane IDs are extracted to detect lane changes. Here mandatory and discretionary lane changes are distinguished with the known structure of the study area. Then corresponding the interacting objective vehicle ID is extracted. The corresponding periods, coexisting starting frames, and scene lasting time of a lane-change scene are extracted to a file as scene parameters. The extracted noisy scene trajectory data are smoothened as in [10] to uncover the personalized and interactive human driving behaviors.

B. Feature Selection

In undertaking this intricate driving endeavor, we implement functional features that prioritize efficiency (such as speed v_x

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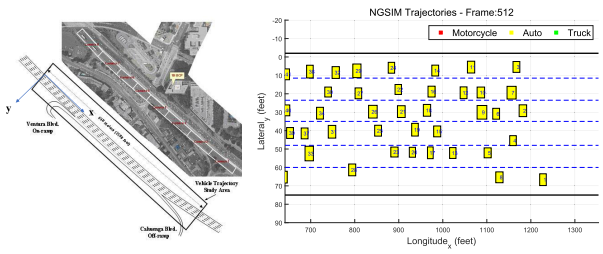

(a) (b)

Fig. 6. The (a) Illustration of the dataset and road structure: The study area on the US Highway 101 with a reconstructed local coordinate system. (b) reconstructed trajectories for scene extracting.

for advancing and successfully merging into lanes), comfort, and energy conservation (discouraging sudden accelerations a_x, a_y or abrupt jerks $\dddot{a}_x$), as well as safety (penalizing time headways in front and behind, THWF and THWB, as well as collisions) [10]. Besides, there is a penalty of general social influence other than the interacting object. It is defined as the anticipated deceleration of surrounding vehicles impacted by the actions of the ego vehicle based on our environmental model. This is expressed as the sum of the absolute values of negative accelerations of the influenced vehicles.

$$f_{GI}(\mathbf{s}_t) = \sum_i |a_i(t)|, \text{ if } a_i(t) < 0, \tag{29}$$

where the variable $a_i(t)$ denotes the acceleration of vehicle i, which is influenced by the actions of the ego vehicle. To avoid repetition, here the interacting object is excluded.

This feature is determined with precision by utilizing a prediction module that forecasts the actions of other agents based on the planned movements of the ego vehicle. By employing driving models such as IDM, this module can accurately predict the behavior of other agents on the road.

In addition, a mandatory lane-changing utility is designed to describe its urgency.

$$f_{MLC}(\mathbf{s}_t) = \frac{L_{C_{tmn}} - x_E}{L_{T_{tmn}}}, \tag{30}$$

where the $L_{T_{tmn}}$ and $L_{C_{tmn}}$ denote the terminal position of the target and current lane, and the x_E denote the current position. The closer the ego is to the merging area, the more advantageous it becomes to switch to the target lane.

Every feature mentioned above is computed at each timestep and aggregated over time to capture the trajectory's characteristics. Subsequently, these trajectory features undergo normalization to ensure consistency, with values falling within the range of 0 to 1. This normalization process involves dividing each feature by the training set's maximum value, thereby normalizing disparate units and scales. Moreover, a constant weight of -10 is assigned to the collision feature. This deliberate imposition of a fixed negative weight is intended to enhance the safety of the model.

C. Implementation Details

The time it takes to finish the learning process (Algorithm 1) with 35 human demonstration trajectories from a vehicle and corresponding $35 * 33 * 33 = 38115$ generated trajectories is roughly 40 minutes with an Intel® Xeon® Processor E5-2678 v3 CPU. While documenting the functional features of all trajectories may initially demand a time investment of 2 hours, this process can be conducted offline and completed in a single session, thereby minimizing any potential inconvenience. While parallel sampling is not the primary focus of this paper, it can greatly expedite the processes of learning and testing.

VII. Results

A. Training Results

Interactive joint trajectories are subtracted from the NGSIM dataset, and each is evenly partitioned into 50 scenes to calibrate a personalized socially compatible driver model. The original lane changing or lane changing attempt process is mostly around 15 seconds in time length, while each scene extracted consists of 3-second-length planning. These scenes boil down to an open-loop rolling-horizon planning process. Among them, 35 are chosen as training data, and the rest 15 scenes form the testing set.

Fig. 7 shows the training process with the average feature difference for three weights and log-likelihood, the weight learning process, and the human likelihood.

Fig. 7a describes the learning process of $\boldsymbol{\theta_{sc}}$

$$\boldsymbol{\theta_{sc}} = \left[\theta_{\text{SI}}, \theta_{\text{SR}}, \theta_{\text{courtesy}}, \theta_{\text{beneficence}}\right] \tag{31}$$

where the first two weights in this framework signify egoistic motivations of self-interest and self-respect, while the last two weights denote altruistic considerations of utility. They are initialized by drawing samples from a normal distribution whose center is 0 and the variance is 0.05. The L2-norm of the social compatibility weight $\boldsymbol{\theta_{sc}}$ increases as the learning process progresses, reducing the randomness in the softmax decision process in Eq. (4). Additionally, they continue to increase the partial derivatives of the log-likelihood concerning $\boldsymbol{\theta_1}$ and $\boldsymbol{\theta_2}$ as shown in Eq. (24). This may obscure the advancement of the functional weights during the learning process. Therefore, we utilize the normalized functional feature norm difference between the ego and object vehicle, $\|\nabla_{\boldsymbol{\theta_1}}\mathcal{J}\|/\|\boldsymbol{\theta_{sc}}\|$, and $\|\nabla_{\boldsymbol{\theta_2}}\mathcal{J}\|/\|\boldsymbol{\theta_{sc}}\|$ to eliminate the effect. Substituting Eq. (4) into them, the weighted differences between the expected features of chosen trajectories and corresponding human demonstrations are obtained.

Figure 7b demonstrates that the three feature norms steadily reduce to a small number. This convergence of the log-likelihood, suggests a higher probability of selecting trajectories closely resembling human driving behavior through the implementation of the proposed SC-MIRL method. The findings demonstrate the effectiveness of the proposed algorithm when coupled with context-aware trajectory sampling. In assessing human likeness, the Best-of-3 Final Displacement Error (FDE) metric is incorporated, showing a gradual convergence in Fig. 7c.

Last but not least, the trained weights of vehicles with IDs of 35 and 33 in NGSIM us-101 with SC-MIRL are listed in Table I. The velocity weight and the mandatory lane change weight are identified to be higher with SC-MIRL than MIRL because of the courtesy and beneficence

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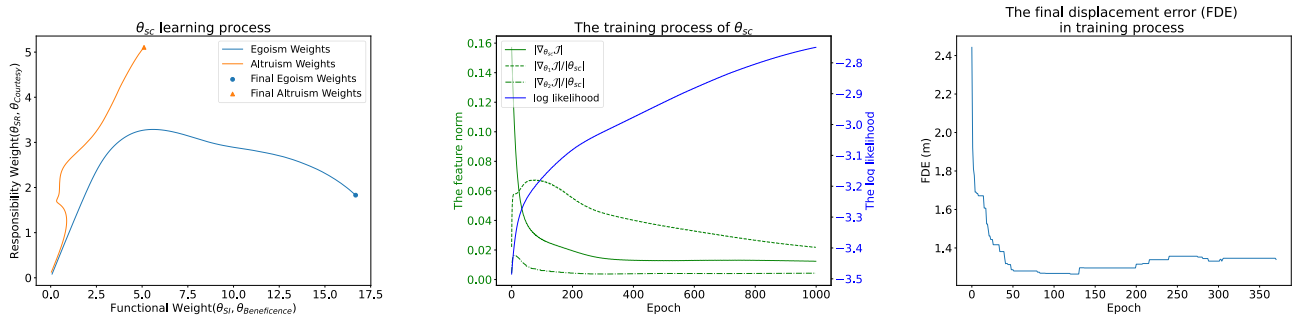

(a) (b) (c)

Fig. 7. Training process analysis. (a) the plot of the $\boldsymbol{\theta_{sc}}$ learning process, with the orange line as the egoism weights, the blue line as the altruism weights, and the two dimensions as functional, and reciprocal responsibility utility weight. (b) plot of the average feature difference for three weights and log-likelihood. (c) plot of FDE training results as the average human likeness.

TABLE I

WEIGHTS OF VEHICLE 35 $\boldsymbol{\theta_1}$ AND VEHICLE 33 $\boldsymbol{\theta_2}$ CALIBRATED BY SC-MIRL AND MIRL

weights	θ_v	θ_{a_x}	θ_{a_y}	θ_{jerk}	θ_{THWF}	θ_{THWB}	θ_{colli}	$\theta_{\text{Social Impact}}$	θ_{md}
$\boldsymbol{\theta_{1-sc}}$	**10.449**	-6.305	-7.369	-8.954	-3.247	1.945	-10.	5.277	**8.576**
$\boldsymbol{\theta_{1-mirl}}$	**0.094**	-0.913	-1.243	-2.595	0.667	-4.905	-10.	0.599	**6.326**
$\boldsymbol{\theta_{2-sc}}$	-6.391	-6.413	-2.61	-6.33	-7.244	-1.651	-10.	-8.183	/
$\boldsymbol{\theta_{2-mirl}}$	-3.262	-3.205	-5.339	-3.062	-4.156	1.564	-10.	-4.9	/

rewards. In terms of safety, comfort, and social impact, it is beneficial for the weights of factors other than velocity (θ_2) to be negative. However, the weight of velocity could also be negative due to personal preference at the current speed level or the Stackelberg Game assumption of the interaction. To improve this further, future studies could explore the use of the Bayesian Game approach.

B. Case Analysis

1) Reciprocal Utility for Social Compatible Choice: We select a series of lane-changing scenes and subtract associated driving trajectories to reproduce the interactions and possible outcomes. The proposed method is adopted to infer the functional rewards of interacting vehicles and the social compatibility rewards of ego vehicles. The learned rewards are then used to interpret the interacting actions.

Figure 8 shows the lateral distance and speed curve of vehicle 35 lane-changing to the front of vehicle 33 in NGSIM scene us-101. The calibrated model of the ego vehicle 35 correctly changes lanes as soon as its target point crosses the lane line at 11s. The safety factor changes from lane-changing condition C_{LC} to car-following condition C_{CF} at 12.5s when the lane line is crossed as shown in Fig. 9.

The most prominent effect of the proposed SC-MIRL is that its resulting driver model decelerates exactly like the demonstration at the end of lane-changing. This phenomenon can be attributed to the reciprocal rewards and the corresponding niches.

The candidate trajectories and the ground-truth human driving trajectories of the interacting vehicles at 11.6s are displayed in Fig. 10a, as well as the trajectories of the two preceding vehicles as the interaction context.

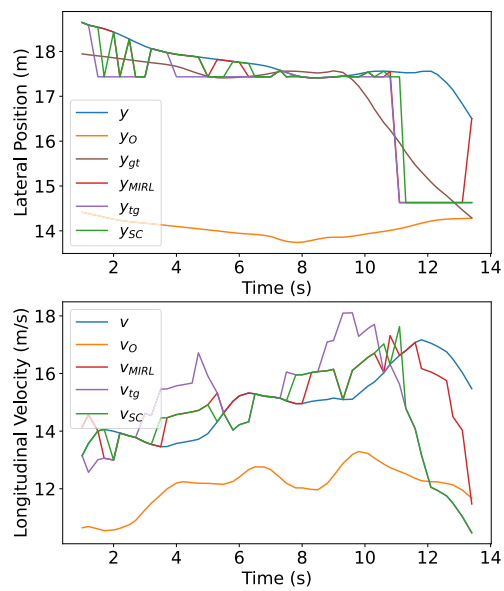

Fig. 8. The comparison results of the calibrated model behavior for vehicle 35 with SC-MIRL and MIRL. The lateral position curve and speed curve of the interaction process. y, v denote the lateral position curve and the velocity curve, respectively. y_O, v_O are the object curves, y_{gt} is the ground truth of y in 3 seconds, while the y_{tg}, v_{tg} are the target value in the proposed sampling. y_{MIRL}, v_{MIRL} and y_{SC}, v_{SC} are the calibration results of the model behavior from MIRL and SC-MIRL.

The algorithm proposed in this study selects the trajectory that more closely aligns with the actual ground truth. Within the lattice of sampled trajectories, the most advantageous choice of SC-MIRL involves conducting a lane change while simultaneously decreasing speed by 3m/s in comparison to the initial speed v_0 measured at 11.6s, 17.1m/s ($v_0 - 3m/s$). Meanwhile, the optimal trajectory for MIRL necessitates executing a lane change with a constant speed of v_0.

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TABLE II
WEIGHTED FUNCTIONAL REWARDS OF VEHICLE 35 AT SCENE 11.6S PRESUMING TO TAKE THE ACTIONS CHOSEN BY SC-MIRL AND MIRL

algo	traj	R_v	R_{a_x}	R_{a_y}	R_{jerk}	R_{THWF}	R_{THWB}	R_{colli}	$R_{\text{Social Impact}}$	R_{md}	p
SC-MIRL	LC($v_0 - 3m/s$)	0.078	-0.497	-0.693	-1.212	0.203	-1.26	-0.	0.074	2.761	0.023
MIRL	LC(v_0)	0.084	**-0.101**	-0.715	**-0.474**	0.278	-1.12	-0.	0.072	2.726	0.057

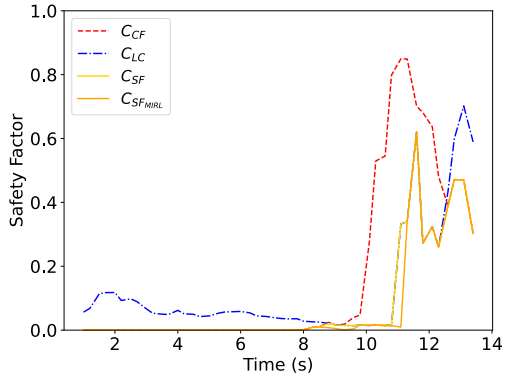

Fig. 9. Plot of environmental niche for vehicle 35.

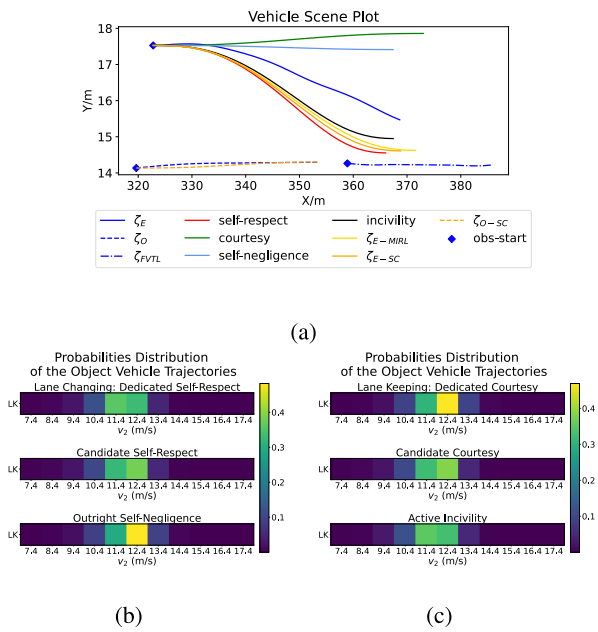

Fig. 10. The possibility distributions over the actions of the object vehicle given actions of the ego at 11.6s. (a) The scene that starts at 11.6s. obs-start denotes the observation at 11.6s. The blue lines denote the ground truth of the ego vehicle ζ_E, the object ζ_O, and the preceding vehicle in the target lane (FVTL). The calibration results of the ego and the object from the SC-MIRL are denoted by ζ_{E-SC} and ζ_{O-SC}, respectively. (b)(c) Demonstrating the object behavior distribution under different ego actions at 11.6s, which determines the self-respect accountability. The possible actions of the ego vehicle include dedicated self-respect and courtesy responsibility, outright self-negligence, active incivility, and a candidate. The candidate here is the SC-MIRL calibrated result.

The MIRL chooses this because of a higher cost of longitudinal acceleration and jerk according to the weighted functional rewards of Table II. Meanwhile, despite a lower self-interest reward, the SC-MIRL chooses to change lanes at the speed of $v_0 - 3m/s$ with a possibility dominance because of the higher self-respect reward according to Table III.

TABLE III
WEIGHTED SOCIAL COMPATIBILITY REWARDS OF VEHICLE 35 AT SCENE 11.6S PRESUMING TO TAKE THE ACTIONS CHOSEN BY SC-MIRL AND MIRL

algo	traj	R_{SI}	R_{SC}	$R_{E_{\text{court}}}$	$R_{E_{\text{benef}}}$	p
SC	LC($v_0 - 3$)	8.794	**17.559**	5.702	-1.012	**0.914**
MIRL	LC(v_0)	10.139	11.020	5.700	-1.135	0.084

Figure 10a illustrates the ego vehicle trajectories corresponding to various levels of reciprocal rewards. These trajectories include the dedicated self-respect responsibility, which is the obligation (lane-changing with $v_0 - 5m/s$), the scenario of outright self-negligence characterized by the lowest self-respect (lane-keeping with $v_0 - 4m/s$), the trajectory most similar to the real scenario (lane-changing with $v_0 - 3m/s$), the courtesy obligation (lane-keeping with v_0), and the instance of active incivility displaying the least courtesy (lane-changing with $v_0 - 4m/s$). The self-respect obligation is chosen since the preceding vehicle in the target lane is slower. It increases its velocity to barely 9.63 m/s in this 3s scene while the ego vehicle drops to 12.1 m/s from the v_0 of 17.1 m/s.

The original speed of the object vehicle is 12.4 m/s. Figure 10b shows that the possibility of keeping it in the next 3 seconds decreases with increased self-respect, while Fig. 10c demonstrates that it increases with more courtesy.

The self-respect reward of deceleration is higher mainly because of a higher environmental niche. On one hand, the self-respect accountability of the result from SC-MIRL towards the ego itself is 0.014, which is less than 0.041, that of the result from MIRL. However, these two trajectories both have little influence on the object vehicle. On the other hand, lane changing with v_0 leads to $C_{CF} = 0.31, C_{LC} = 0.44$ as shown in 9 while lane changing with $v_0 - 3m/s$ leads to $C_{CF} = 0.70, C_{LC} = 0.62$. This is because the lower target velocity gives a bigger car-following gap of 16.3m between the FVTL and the ego vehicle instead of 13.9m with v_0. Besides, larger deceleration induces a larger heading angle, which makes the remaining lane change faster.

Figure 11 shows that the calibrated FDE of the ego vehicle with the proposed SC-MIRL is remarkably lower by up to 5m after 11.1s. The cross-entropy loss is also significantly lower and reaches 0.014. Actually, ego vehicle crosses the lane line 3 seconds later. Therefore, the self-respect obligation trajectory is chosen to prepare for the car-following behavior in the target lane. Furthermore, this increases the probability of lane-changing action as is shown in Table III, decreases the uncertainty of the ego behavior and therefore decreases its cross-entropy.

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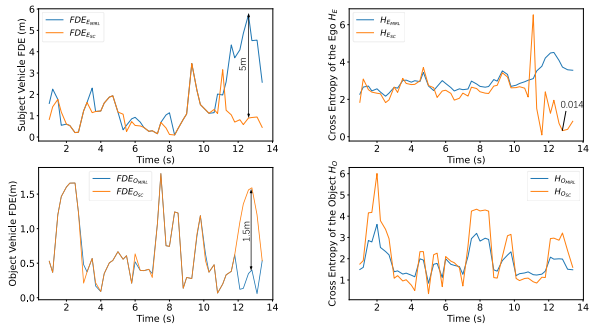


(a) (b)

Fig. 11. (a) The best-of-3 FDE of the ego vehicle and the object calibrated by SC-MIRL and MIRL. (b) Demonstrating the cross entropy loss of the vehicle 35 and 33 calibrated by SC-MIRL and MIRL.

Meanwhile, it is worth noting that the FDE of the object is also bigger by up to 1.5m. The same goes for the cross-entropy of the object. This phenomenon indicates that the ego's first-order false belief of the object, represented by $\boldsymbol{\theta_2}$, is recognized, explaining the abrupt slowdown of the ego's movement after the 11th second as shown in Figure 8.

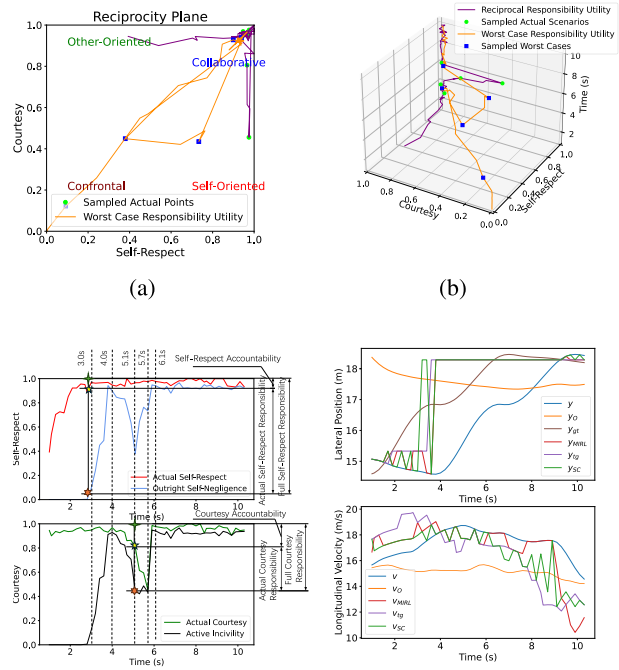
This case demonstrates the effectiveness of the proposed reciprocal rewards, conditions, and the calibration algorithm of SC-MIRL.

2) SR-C Plane for Social Interaction Situation: An examination of a mandatory lane-changing procedure that impacts both interacting vehicles is conducted utilizing the proposed reciprocity plane model. The actual reciprocal responsibility utility and worst-case scenarios are depicted in Fig. 12a while their evolution process in time is shown in Fig. 12b and Fig. 12c. During this process, four scenes of importance are sampled and analyzed. They are the scenes at 3.0s, 4.0s, 5.1s, 5.7s, and 6.1s.

Figure 12a and Figure 12b show that the reciprocal responsibility utility begins with an other-oriented tendency and then goes to collaborative interaction in the fourth second. This could be due to its acceleration to change lanes in 3 seconds as shown in Fig. 12d. Because of the lane change, the action of the ego becomes suddenly subject-oriented at the fifth second only to go back and converge to be collaborative after 6 seconds. The worst situation also goes from being confrontational to collaborative before the fourth second. It returns to confrontational (0.4, 0.4) at 5.1s, becomes a bit more self-respect at 5.7s, and finally converges to collaborative interaction.

Two dimensions of the reciprocity plane, the self-respect and courtesy responsibilities are shown in Fig. 12c. The accountability, full responsibility, and actual responsibility of self-respect at 3s and those of courtesy at 5.1s are especially depicted here.

Meanwhile, the worst-case responsibility indicates the influence capacity of the interaction and the lane-changing conditions. The worst-case responsibilities (active incivility and outright self-negligence) first remain at 0 and then increase to 0.9 from 3s to 4s. This indicates that the influence capacity of the interaction decreases and the lane-changing conditions



(a) (b) (c) (d)

Fig. 12. The interaction intensity and tendency analysis of vehicle 631 with SR-C plane. (a) The reciprocity state trajectory and the reciprocity plane. (b) The reciprocity state trajectory in time. (c) The responsibilities of both parties. (d) The behavior of vehicles interacting during the lane-change process, including the lateral position curve and speed curve.

improve because of the acceleration during this time. Therefore, the calibrated result of SC-MIRL chooses to change lanes then, which decreases the actual courtesy.

The worst-case responsibilities also decrease until 5.1s. At 5.1s, the ego will collide with its preceding vehicle at the original lane because of the high longitudinal speed for lane-changing. This situation changes after 5.1s because of its lateral movement and improves the outright self-negligence situation.

Furthermore, after 5.7s, the ego will not only avoid colliding with the preceding vehicles, but also steer clear of colliding with the object vehicle in each sampled trajectory. This improves the active incivility situation.

The SR-C plane also converges to collaboration when the full responsibility is around 0.1 and the actual responsibility is less.

This case demonstrates that the proposed SR-C plane can clearly describe the bilateral influence of ego actions through the lens of accountability and responsibility.

C. Statistical Analysis

This study focuses on 38 carefully selected lane-change scenarios and examines the key interactions between the ego vehicle and the designated object vehicle. These scenarios are categorized into two groups: discretionary lane changes (15 instances) and mandatory lane changes (23 instances). They are adopted to calibrate the personalized driver model with both MIRL and the proposed SC-MIRL. Figure 13 depicts

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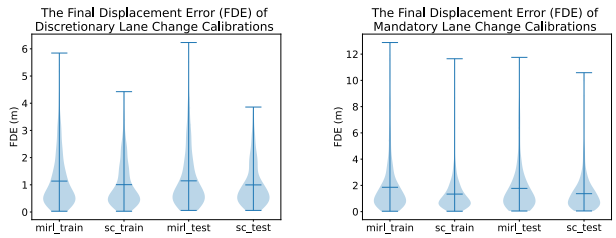

(a) (b)

Fig. 13. SC-MIRL calibration statistical results compared to MIRL. (a) discretionary Lane-Change. (b) mandatory Lane-Change.

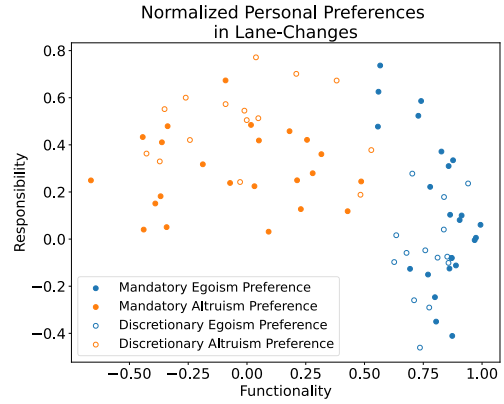

Fig. 14. The calibrated results of the personal social compatibility reward weights $\boldsymbol{\theta_{sc}}$. The Egoism utility parameters encompass the weight for self--interest θ_{SI} (functionality) and self-respect θ_{SR} (responsibility), while the altruism utility parameters refer to the weights of beneficence $\theta_{beneficence}$ (functionality) and courtesy $\theta_{courtesy}$ (responsibility).

the calibration results. The discretionary lane-changing results are shown in Fig. 13a. The mean FDE in 3 seconds of the SC-MIRL training results is 1.006m, and that of the testing results is 0.997m, which improves the training and testing results of the MIRL by 11.522% and 13.002%, respectively. The maximum of the SC-MIRL calibration error is 4.425m, while that of MIRL is 5.845m.

Figure 11b shows that the enhancement in mandatory lane-changing is significantly more noticeable due to the heightened intensity of interaction. The training results from SC-MIRL are improved by 27.948% while the testing results improved by 22.743%. The mean of the training results error is 1.338m while that of the testing results is 1.369m, which indicates the good generalization ability of the model. The maximum of the SC-MIRL calibration error is 11.648m, which is also smaller than that of the MIRL, 12.883m. However, these are bigger than that of the discretionary lane changes since the interactions are more intense and the speed fluctuates more. Moreover, the vehicle with ID 97 on US Highway 101 engaged in a collision with the MIRL due to the MIRL calibration results inaccurately indicating a lane change ahead, without considering proper courtesy.

In conclusion, the proposed SC-MIRL demonstrates a clear advantage over the MIRL baseline in terms of human driver calibration, particularly within mandatory lane-changing situations where interactions are more confrontational in nature.

The calibrated $\boldsymbol{\theta_{sc}}$ of the discretionary and mandatory lane-changing scenarios are shown in Fig. 14.

There are three potential conclusions that can be drawn.

1) The minimum of the θ_{SI} is bigger than the maximum of the $\theta_{beneficence}$. While all the θ_{SI} are positive, the $\theta_{beneficence}$ can be negative. The variance of the latter is also bigger. This result can be interpreted as the inherent tendency for individuals to prioritize self-interest over the interests of others, with the degree of concern for others varying among individuals.
2) All the $\theta_{courtesy}$ are positive, which indicates the significance of the accountability to others to the human drivers. The mean of the $\theta_{courtesy}$ of the discretionary results, 0.490 is bigger than that of the mandatory scenarios, 0.280. This means that drivers place a higher importance on courtesy in discretionary situations where interactions are less confrontational.
3) The mean of the $\theta_{self-respect}$ of the mandatory results, 0.112, is bigger than that of the discretionary scenarios, -0.053. Meanwhile, the variance of the $\theta_{self-respect}$ in the mandatory scenarios, 0.113, is also bigger than that of the discretionary cases, 0.035. The results suggest that individuals with higher levels of self-respect may be more likely to participate in confrontational situations that are required, while also showing more variance in their adherence to mandatory lane-changing plans.

D. Ablation Experiments

To analyze the importance of the social-compatibility utilities and niches, an ablation study for these four features is conducted in Tab. IV and Tab. V. The contributions of the four components and two niches are examined by dropping one of them at a time.

All the social compatibility features and niches contribute to improving human likeness since the complete SC-MIRL model gets the lowest FDE and cross-entropy. The model without a safety factor shows a slightly lower training FDE of 0.998m in mandatory cases. However, it also displays a higher testing FDE of 1.015m compared to the proposed SC-MIRL model. On the other hand, discarding beneficence in mandatory lane changes may decrease the testing FDE at the cost of a higher cross-entropy loss.

In analyzing the four social compatibility rewards, it is evident that self-interest rewards play a pivotal role in both scenarios, showing the highest FDEs and cross-entropy when the self-interest reward is absent. Following closely behind is the courtesy reward, which holds significant importance in discretionary cases. Conversely, self-respect becomes more prominent in testing situations involving mandatory lane changes. It is worth noting that beneficence ranks lowest in impact in mandatory lane-change scenarios.

The incentive factor plays a more significant role than the safety factor in discretionary testing scenarios. On the other hand, the safety factor takes precedence in mandatory lane changes. This phenomenon can be attributed to the fact that incentives primarily drive discretionary lane changes, whereas safety considerations take precedence in mandatory lane changes.

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TABLE IV

THE ABLATION STUDY OF SOCIAL COMPATIBILITY UTILITIES WITH DISCRETIONARY LANE-CHANGING SCENARIOS. H_{dis} IS THE CROSS ENTROPY LOSS OF THE DISCRETIONARY LANE-CHANGE SCENARIOS

Model	FDE_{dis-tr}	FDE_{dis-te}	H_{dis}
MIRL	1.137	1.146	2.499
SC-MIRL	1.006	**0.997**	**2.251**
W/o Self-Interest	1.768	2.013	2.924
W/o Self-Respect	1.023	1.028	2.264
W/o Courtesy	1.12	1.151	2.436
W/o Beneficence	1.007	1.005	2.255
W/o Safety Factor	**0.998**	1.015	2.273
W/o Incentive Factor	1.05	1.126	2.385

TABLE V

THE ABLATION STUDY OF SOCIAL COMPATIBILITY UTILITIES WITH MANDATORY LANE-CHANGING SCENARIOS. H_{md} IS THE CROSS ENTROPY LOSS OF THE MANDATORY LANE-CHANGE SCENARIOS

Model	FDE_{md-tr}	FDE_{md-te}	H_{md}
MIRL	1.857	1.772	3.044
SC-MIRL	**1.338**	1.369	**2.569**
W/o Self-Interest	2.08	1.861	3.132
W/o Self-Respect	1.395	1.425	2.647
W/o Courtesy	1.542	1.482	2.687
W/o Beneficence	1.339	**1.309**	2.621
W/o Safety Factor	1.392	1.452	2.638
W/o Incentive Factor	1.448	1.444	2.669

VIII. DISCUSSIONS

We have designed a responsibility-based socially compatible lane-changing behavior model which embraces the responsibilities of self-respect and courtesy. It combines prosocial functional utility, reciprocal responsibility utilities, and environmental niches. The proposed model also adopts adaptive environmental niches of safety and incentive factors to evaluate the value of action results. To verify the proposed model, we design a hierarchical SC-MIRL algorithm and systematically adapt a multi-agent simulation environment based on the NGSIM dataset. We subtract mandatory and discretionary lane changes from the dataset for calibration. Besides, by incorporating these two responsibilities into an SR-C plane, we can evaluate the intensity and direction of our actions. This research integrates causality and reachability to calculate responsibility, utilizing a regret-based model. Accountability is examined concerning an ideal obligation for every participant in interaction scenarios. One could leverage Nash equilibrium to optimize principles and extend the suggested approach for navigating AVs in diverse interactive situations involving uncertain behavior and conflicting interests. Additionally, the niche model facilitates the adaptive preference model to unfreeze AVs safely. Moreover, incorporating acceleration to the conflict zone as thresholds outlined in Equation 18 is instrumental in modeling the intersection interaction niche. Employing a learning-based niche model offers an effective strategy for accommodating a wide range of interactions in various contexts [62]. Therefore, this study not only contributes to responsible and socially compatible autonomous driving interactive decision-making, but also gives insight into stable multi-agent reinforcement learning [63] and trust behaviors of large language models (LLMs), namely the behavioral factors of risk perception and prosocial preference [64].

Utilizing an SC-MIRL algorithm to calibrate the proposed model, we observe a reduction in behavioral uncertainty through responsible decision-making. To be specific, the proposed SC-MIRL considers lane change in advance to carve its safety niches and improve the actual self-respect responsibility rather than solely focusing on acceleration or jerks at crucial moments. Therefore, the phenomenon emerges that a proactive approach allows for smoother transitions in response to traffic conditions in the adjacent lane, reducing behavioral uncertainty. The cross-entropy can then be reduced to 0.09 in case analysis, where the probability of making the right choice is up to 0.914. The average FDE in 3s is reduced by 22.743% for mandatory lane changes in statistical analysis. When comparing calibrated weights, it is evident that individuals typically prioritize their own needs in terms of functional utility preference. However, the level of concern for others may vary among different individuals. In terms of responsibility preference, people value courtesy more in discretionary cases, meanwhile, greater average and variance are found in mandatory cases. Ablation study shows that all parts of the model contribute to the accuracy and generalization ability, while they play different roles in lane-change types. Courtesy and incentive factors are concerned more with discretionary lane changes, while safety issues and self-respect weigh heavier in mandatory lane changes.

The proposed method can help unfreeze autonomous driving, evaluate interaction intensity and tendency, and provide insight into responsibility in an interactive environment with behavioral uncertainty. Future developments may involve the incorporation of learning-based multi-modal sampling techniques and socially compatible multi-agent interactions within complex road infrastructures.

APPENDIX
SC-MIRL PARAMETERS UPDATING

In this section, we provide a comprehensive breakdown of the SC-MIRL parameters iteration formula along with the pertinent parameter values, allowing readers to implement it in their research. First, we illustrate The partial derivative of social compatibility features with respect to functional parameters $\nabla_{\boldsymbol{\theta}_1}\mathbf{F}$, $\nabla_{\boldsymbol{\theta}_2}\mathbf{F}$ in (24).

$$\nabla_{\boldsymbol{\theta}_1}\mathbf{F} = \left[\sum_{j=1}^{M}\mathbf{f}_{\tilde{\zeta}^i}P_{\tilde{\zeta}^j}, \mathbf{0}, \mathbf{0}, \mathbf{0}\right] \tag{32}$$

$$\nabla_{\boldsymbol{\theta}_2}\mathbf{F} = \Big[\nabla_{\boldsymbol{\theta}_2}R_{E_{\mathrm{SI}}}\left(\mathbf{x}^0, \zeta_E\right), \nabla_{\boldsymbol{\theta}_2}R_{E_{\mathrm{SR}}}\left(\mathbf{x}^0, \zeta_E\right), \nabla_{\boldsymbol{\theta}_2}R_{E_{\mathrm{cour}}}\left(\mathbf{x}^0, \zeta_E\right), \nabla_{\boldsymbol{\theta}_2}R_{E_{\mathrm{benef}}}\left(\mathbf{x}^0, \zeta_E\right)\Big] \tag{33}$$

where

$$\nabla_{\boldsymbol{\theta}_2}R_{E_{\mathrm{SI}}}\left(\mathbf{x}^0, \zeta_E\right) = \sum_{j=1}^{M}\boldsymbol{\theta}_1{}^T\mathbf{f}_{\tilde{\zeta}^i}\nabla_{\boldsymbol{\theta}_2}P_{\tilde{\zeta}^j}, \tag{34}$$

$$\nabla_{\boldsymbol{\theta}_2}R_{E_{\mathrm{SR}}}\left(\mathbf{x}^0, \zeta_E\right) = R_{E_{SR}}\sum_{j=1}^{M}\Big(\nabla_{\boldsymbol{\theta}_2}P_{\tilde{\zeta}^j}C_{SF}^{j}$$

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$$- \nabla_{\boldsymbol{\theta_2}} \left(P_{\tilde{\zeta}_{SO}^j} \log_2 \frac{P_{\tilde{\zeta}_{SO}^j}}{P_{\tilde{\zeta}^j}} \right) \Bigg), \tag{35}$$

$$\nabla_{\boldsymbol{\theta_2}} R_{E_{\text{cour}}} \left(\mathbf{x}^0, \zeta_E \right) = R_{E_{\text{court}}} \sum_{j=1}^{M} \Bigg(\nabla_{\boldsymbol{\theta_2}} P_{\tilde{\zeta}^j} C_{IF}^j - \nabla_{\boldsymbol{\theta_2}} \left(P_{\tilde{\zeta}_{CO}^j} \log_2 \frac{P_{\tilde{\zeta}_{CO}^j}}{P_{\tilde{\zeta}^j}} \right) \Bigg), \tag{36}$$

$$\nabla_{\boldsymbol{\theta_2}} R_{E_{\text{benef}}} \left(\mathbf{x}^0, \zeta_E \right) = \sum_{j=1}^{M} \mathbf{f}_{\tilde{\zeta}^j} P_{\tilde{\zeta}^j} + \sum_{j=1}^{M} \boldsymbol{\theta_2}^T \mathbf{f}_{\tilde{\zeta}^j} \nabla_{\boldsymbol{\theta_2}} P_{\tilde{\zeta}^j} \tag{37}$$

where

$$\nabla_{\boldsymbol{\theta_2}} \left(P_{\tilde{\zeta}_{SO}^j} \log_2 \frac{P_{\tilde{\zeta}_{SO}^j}}{P_{\tilde{\zeta}^j}} \right) = \nabla_{\boldsymbol{\theta_2}} P_{\tilde{\zeta}_{SO}^j} \log \frac{P_{\tilde{\zeta}_{SO}^j}}{P_{\tilde{\zeta}^j}} + \frac{\nabla_{\boldsymbol{\theta_2}} P_{\tilde{\zeta}_{SO}^j}}{\log 2} - \frac{1}{\log 2} \frac{P_{\tilde{\zeta}_{SO}^j}}{P_{\tilde{\zeta}^j}} \nabla_{\boldsymbol{\theta_2}} P_{\tilde{\zeta}^j} \tag{38}$$

$$\nabla_{\boldsymbol{\theta_2}} \left(P_{\tilde{\zeta}_{CO}^j} \log_2 \frac{P_{\tilde{\zeta}_{CO}^j}}{P_{\tilde{\zeta}^j}} \right) = \nabla_{\boldsymbol{\theta_2}} P_{\tilde{\zeta}_{CO}^j} \log \frac{P_{\tilde{\zeta}_{CO}^j}}{P_{\tilde{\zeta}^j}} + \frac{\nabla_{\boldsymbol{\theta_2}} P_{\tilde{\zeta}_{CO}^j}}{\log 2} - \frac{1}{\log 2} \frac{P_{\tilde{\zeta}_{CO}^j}}{P_{\tilde{\zeta}^j}} \nabla_{\boldsymbol{\theta_2}} P_{\tilde{\zeta}^j} \tag{39}$$

Considering Eq. (7), the partial derivatives of probability are:

$$\nabla_{\boldsymbol{\theta_2}} P_{\tilde{\zeta}^j} = P_{\tilde{\zeta}^j} \frac{\mathbf{f}_{\tilde{\zeta}^j} \sum_{k=1}^{M} e^{\boldsymbol{\theta_2}^T \mathbf{f}_{\tilde{\zeta}^k}} - \sum_{k=1}^{M} e^{\boldsymbol{\theta_2}^T \mathbf{f}_{\tilde{\zeta}^k}} \mathbf{f}_{\tilde{\zeta}^k}}{\sum_{k=1}^{M} e^{\boldsymbol{\theta_2}^T \mathbf{f}_{\tilde{\zeta}^k}}} \tag{40}$$

$$\nabla_{\boldsymbol{\theta_2}} P_{\tilde{\zeta}_{SO}^j} = P_{\tilde{\zeta}_{SO}^j} \frac{\mathbf{f}_{\tilde{\zeta}_{SO}^j} \sum_{k=1}^{M} e^{\boldsymbol{\theta_2}^T \mathbf{f}_{\tilde{\zeta}^k}} - \sum_{k=1}^{M} e^{\boldsymbol{\theta_2}^T \mathbf{f}_{\tilde{\zeta}^k}} \mathbf{f}_{\tilde{\zeta}^k}}{\sum_{k=1}^{M} e^{\boldsymbol{\theta_2}^T \mathbf{f}_{\tilde{\zeta}^k}}} \tag{41}$$

$$\nabla_{\boldsymbol{\theta_2}} P_{\tilde{\zeta}_{CO}^j} = P_{\tilde{\zeta}_{CO}^j} \frac{\mathbf{f}_{\tilde{\zeta}_{CO}^j} \sum_{k=1}^{M} e^{\boldsymbol{\theta_2}^T \mathbf{f}_{\tilde{\zeta}^k}} - \sum_{k=1}^{M} e^{\boldsymbol{\theta_2}^T \mathbf{f}_{\tilde{\zeta}^k}} \mathbf{f}_{\tilde{\zeta}^k}}{\sum_{k=1}^{M} e^{\boldsymbol{\theta_2}^T \mathbf{f}_{\tilde{\zeta}^k}}} \tag{42}$$

Besides, utilizing the gradient ascent method, we can continuously adjust the hierarchical weights until convergence. We incorporate L2 regularization on the weights in the objective function to avoid overfitting.

$$\begin{aligned} \nabla_{\boldsymbol{\theta_{sc}}} \mathcal{J} &= \nabla_{\boldsymbol{\theta_{sc}}} \mathcal{J} - 2\lambda \boldsymbol{\theta_{sc}} \\ \nabla_{\boldsymbol{\theta_1}} \mathcal{J} &= \nabla_{\boldsymbol{\theta_1}} \mathcal{J} - 2\lambda \boldsymbol{\theta_1} \\ \nabla_{\boldsymbol{\theta_2}} \mathcal{J} &= \nabla_{\boldsymbol{\theta_2}} \mathcal{J} - 2\lambda \boldsymbol{\theta_2} \end{aligned} \tag{43}$$

where $\lambda = 0.01$ is the regularization parameter.

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Tingjun Li received the B.E. degree in automobile service engineering from Jilin University, Changchun, China, in 2019. He is currently pursuing the Ph.D. degree in automotive engineering with Jilin University, Changchun. His current research focuses on interactive decision-making for socially compatible and trustworthy autonomous driving.

Nan Xu received the Ph.D. degree from Jilin University, Changchun, China, in 2012. In 2019, he was a Visiting Scholar with the Department of Mechanical and Mechatronics Engineering, University of Waterloo. He became the Tang Aoqing Professor of Jilin University in 2020. He is currently a Professor with the National Key Laboratory of Automotive Chassis Integration and Bionics, Jilin University of China. His current research focuses on tire dynamics and control, intelligent tire, vehicle state estimation, and dynamics and stability control of electric vehicles and autonomous vehicles.

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Shuo Feng (Member, IEEE) received the bachelor's and Ph.D. degrees from the Department of Automation, Tsinghua University, China, in 2014 and 2019, respectively. He was a Joint Ph.D. Student in civil and environmental engineering with the University of Michigan, Ann Arbor, MI, USA, from 2017 to 2019, where he was a Post-Doctoral Researcher from 2019 to 2021 and an Assistant Research Scientist from 2021 to 2023. He is currently an Associate Professor with the Department of automation, Tsinghua University. His current research interests include connected and automated vehicle evaluation, mixed traffic control, and transportation data analysis.

Hassan Askari was born in Rasht, Iran. He received the Ph.D. degree in mechanical and mechatronics engineering from the University of Waterloo, Waterloo, ON, Canada, in 2019. He has co-authored one book and one book chapter published by Springer. He has authored or co-authored more than 70 journal and conference papers in the areas of nonlinear vibrations, applied mathematics, nanogenerators, and self-powered sensors. He was a recipient of several prestigious awards, including the Outstanding Researcher at Iran University of Science and Technology, the Fellowship of the Waterloo Institute of Nanotechnology, Waterloo, Ontario Graduate Scholarship, and the University of Waterloo President Award. He is an active reviewer of more than 40 journals and is an editorial board member of several scientific and international journals.

Bruno Henrique Groenner Barbosa received the B.Sc. degree in control and automation engineering and the M.Sc. and Ph.D. degrees in electrical engineering from the Federal University of Minas Gerais, Belo Horizonte, Brazil, in 2003, 2006, and 2009, respectively. In 2008, he was a Visiting Researcher with the University of New South Wales, Sydney, NSW, Australia. In 2020, he was a Visiting Professor with the Department of Mechanical and Mechatronics Engineering, University of Waterloo, Waterloo, ON, Canada. He is an Associate Professor with the Department of Automatics, Federal University of Lavras, Lavras, Brazil, where he heads the Artificial Intelligence and Automation Research Group. His research interests include artificial intelligence techniques, such as neural networks and evolutionary algorithms, for solving real-world engineering problems. These have included identification of nonlinear dynamical systems, development of soft sensors, pattern recognition, and system optimization and control.

Konghui Guo received the B.Sc. degree in automobile and tractor from Jilin University of Technology in 1956. He was named a Doctoral Supervisor in 1990 and was elected as an Academician of Chinese Academy of Engineering in 1994. He is currently the Honorary Dean of the School of Automotive Engineering, Jilin University, and a member of the editorial board of IATSS Research and Vehicle System Dynamics. He was the Chief Engineer of Changchun Automobile Research Institute, a Visiting Researcher at the University of Michigan, the Chairperson of the Technical Committee of the 5th Annual Conference of the International Society of Automotive Engineers of the Pacific, and the Chairperson of the Technical Committee of the 25th Annual Conference of the World Society of Automotive Engineers (FISITA).

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